

Electromagnetic Radiation from Cell Towers

A brief discussion and explanation of relevant technical issues is presented here to help readers in better understanding of the effects of electromagnetic radiation from cellphone towers

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Of late, a lot of concern has been voiced in electronic as well as print media about uninhibited installation of cellphone towers in thickly populated areas of many cities in India. There is a perception relating to existence of a high level of electromagnetic radiation in the vicinity of these towers, which may cause adverse biological effects. In the absence of suitable policy directives for cell tower installations, monitoring, effective control and execution at all levels by governmental bodies, such mushrooming of cellphone transmitting towers could create a sense of panic amongst general public.

As the cellphone base in India is expanding at nearly 17 per cent per annum (it may cross 500 million subscribers by the end of this year), the requirement for cellphone towers will also grow proportionately. It is therefore high time that a strict regulatory regime is established as early as possible to avoid possible fallout.

There is a need for not only controlling the haphazard installation of the towers but also undertaking systematic study for measurements of the radiation levels in some selected high-population-density urban areas to ensure that the power density levels are well below the prescribed threshold limits. The results need to be made available for public education and scrutiny. The present threshold limits prescribed by the government are considered to be rather too generous (one of the highest

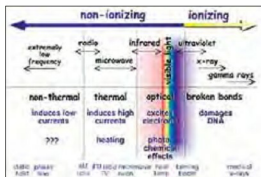


Fig. 1: Spectrum of EM Radiation

TABLE-I
Power Density Conversion
Between Units

Conversion between W/m ² , mW/cm ² and μ W/cm ²		
0.000.01 W/m ²	0.000.001 mW/cm ²	0.001 μ W/cm ²
0.000.1 W/m ²	0.000.01 mW/cm ²	0.01 μ W/cm ²
0.001 W/m ²	0.000.1 mW/cm ²	0.1 μ W/cm ²
0.01 W/m ²	0.001 mW/cm ²	1 μ W/cm ²
0.1 W/m ²	0.01 mW/cm ²	10 μ W/cm ²
1 W/m ²	0.1 mW/cm ²	100 μ W/cm ²
10 W/m ²	1 mW/cm ²	1,000 μ W/cm ²
100 W/m ²	10 mW/cm ²	10,000 μ W/cm ²
1,000 W/m ²	100 mW/cm ²	100,000 μ W/cm ²
10,000 W/m ²	1,000 mW/cm ²	1,000,000 μ W/cm ²
100,000 W/m ²	10,000 mW/cm ²	10,000,000 μ W/cm ²

W/m² = watts per square metre, mW/cm² = milliwatts per square centimetre, μ W/cm² = microwatts per square centimetre

in the world), and hence there is a need to review and remedy the situation and not wait until it becomes the subject matter of a public-interest petition in the light of possible environmental adverse effects.

In the aftermath of the court verdict relating to Bhopal tragedy and the key clause in the pending nuclear liability bill, there is a growing concern about harmful affects of radiation from the

towers. While Cellular Operators Association of India (COAI) maintains that the alleged health hazard from EM radiation from cellphone towers and mobile phones is a misconception, devoid of any underlying basis in reality, however, many others do not share their views and cite the same and other documents to stress their viewpoint. For readers to form their own independent

and informed opinion, a certain amount of technical background and knowledge is necessary. The following text briefly covers technical aspects of the entire matter.

Ionising and non-ionising radiation

Fig. 1 shows different forms of electromagnetic energy in the entire spectrum of EM radiation with the range of frequencies, their energies, effects and sources.

When referring to biologic radiation exposures, EM radiation is divided into two types: ionising and non-ionising. Because the human body is composed of about 60 per cent water, ionising and non-ionising radiations refer to whether the RF energy is high enough to break chemical bonds of water (ionising) or not (non-ionising). Technically, ionising radiation is the amount of energy that can remove an electron from a water molecule (1.216 kilojoules/mol), and non-ionising radiation is anything less than that amount.